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Editorial note. All mathematical statement and proof text is reproduced verbatim from the author's chapter derived.tex of the Stacks Project, fixed at commit a04446e57ec1fbc252a871afcec7752fb2807b14. The pinned preamble is supplied as its own file; the document wrapper class, the xr-hyper wiring and the \externaldocument list are omitted because the assembler supplies the class.

Editorial note. Numbering inside this standalone excerpt is editorial, not the full Stacks book print numbering.

Editorial note. The candidate was previously judged H2 by the initial-corpus stage; that judgement and its reason are reused verbatim and were not re-derived here. Original candidate key: S001-lemma-brown-bis.

Editorial note. Reference labels in the proof resolve to navigation pointers to the original author source; these pointers do not reproduce proofs of the prerequisites.

Allowed standard prerequisites (editorial).

- Prerequisites named by the author inside the counted span. The counted span is reproduced verbatim, including the author's own citations. Its prerequisites are the earlier results of the same fixed Stacks Project version that the author cites there and in the chapter's lead-in.

Source reference. Original lemma 0A8F: sources/derived.tex, source lines 11329–11329. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 001P: sources/categories.tex, source lines 541–541. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 05RD: sources/derived.tex, source lines 1813–1813. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{D}$ be a triangulated category with direct sums. Suppose given a set $\mathcal{E}$ of objects of $\mathcal{D}$ such that*

(1) *if X is a nonzero object of $\mathcal{D}$, then there exists an $E \in \mathcal{E}$ and a nonzero map $E \to X$, and*

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(2) *given objects X_n, $n \in \mathbf{N}$ of $\mathcal{D}$, $E \in \mathcal{E}$, and $\alpha : E \to \bigoplus X_n$, there exist $E_n \in \mathcal{E}$ and $\beta_n : E_n \to X_n$ and a morphism $\gamma : E \to \bigoplus E_n$ such that $\alpha = (\bigoplus \beta_n) \circ \gamma$.*

Let $H : \mathcal{D} \to Ab$ be a contravariant cohomological functor which transforms direct sums into products. Then H is representable.

Proof. This proof is very similar to the proof of Lemma 0A8F. We may replace $\mathcal{E}$ by $\bigcup_{i \in \mathbf{Z}} \mathcal{E}[i]$ and assume that $\mathcal{E}$ is preserved by shifts. Consider pairs (E, a) where $E \in \mathcal{E}$ and $a \in H(E)$ and set

$$X_1 = \bigoplus\nolimits_{(E,a)} E$$

Since $H(X_1) = \prod_{(E,a)} H(E)$ we see that $(a)_{(E,a)}$ defines an element $a_1 \in H(X_1)$. Set $H_1 = \mathrm{Hom}_{\mathcal{D}}(-, X_1)$. By Yoneda's lemma (Categories, Lemma 001P) the element a_1 defines a natural transformation $H_1 \to H$.

We are going to inductively construct X_n and transformations $a_n : H_n \to H$ where $H_n = \mathrm{Hom}_{\mathcal{D}}(-, X_n)$. Namely, we apply the procedure above to the functor $\mathrm{Ker}(H_n \to H)$ to get an object

$$K_{n+1} = \bigoplus\nolimits_{(E,k),\ k \in \mathrm{Ker}(H_n(E) \to H(E))} E$$

and a transformation $\mathrm{Hom}_{\mathcal{D}}(-, K_{n+1}) \to \mathrm{Ker}(H_n \to H)$. By Yoneda's lemma the composition $\mathrm{Hom}_{\mathcal{D}}(-, K_{n+1}) \to H_n$ gives a morphism $K_{n+1} \to X_n$. We choose a distinguished triangle

$$K_{n+1} \to X_n \to X_{n+1} \to K_{n+1}[1]$$

in $\mathcal{D}$. The element $a_n \in H(X_n)$ maps to zero in $H(K_{n+1})$ by construction. Since H is cohomological we can lift it to an element $a_{n+1} \in H(X_{n+1})$.

Set $X = \mathrm{hocolim} X_n$. Applying H to the defining distinguished triangle

$$\bigoplus X_n \to \bigoplus X_n \to X \to \bigoplus X_n[1]$$

we obtain an exact sequence

$$\prod H(X_n) \leftarrow \prod H(X_n) \leftarrow H(X)$$

Thus there exists an element $a \in H(X)$ mapping to (a_n) in $\prod H(X_n)$. Hence a natural transformation $\mathrm{Hom}_{\mathcal{D}}(-, X) \to H$ such that

$$\mathrm{Hom}_{\mathcal{D}}(-, X_1) \to \mathrm{Hom}_{\mathcal{D}}(-, X_2) \to \mathrm{Hom}_{\mathcal{D}}(-, X_3) \to \ldots \to \mathrm{Hom}_{\mathcal{D}}(-, X) \to H$$

commutes. We claim that $\mathrm{Hom}_{\mathcal{D}}(-, X) \to H(-)$ is an isomorphism.

Let $E \in \mathcal{E}$. Let us show that

$$\mathrm{Hom}_{\mathcal{D}}(E, \bigoplus X_n) \to \mathrm{Hom}_{\mathcal{D}}(E, \bigoplus X_n)$$

is injective. Namely, let $\alpha : E \to \bigoplus X_n$. Then by assumption (2) we obtain a factorization $\alpha = (\bigoplus \beta_n) \circ \gamma$. Since $E_n \to X_n \to X_{n+1}$ is zero by construction, we see that the composition $\bigoplus E_n \to \bigoplus X_n \to \bigoplus X_n$ is equal to $\bigoplus \beta_n$. Hence also the composition $E \to \bigoplus X_n \to \bigoplus X_n$ is equal to α. This proves the stated injectivity and hence also

$$\mathrm{Hom}_{\mathcal{D}}(E, \bigoplus X_n[1]) \to \mathrm{Hom}_{\mathcal{D}}(E, \bigoplus X_n[1])$$

is injective. It follows that we have an exact sequence

$$\mathrm{Hom}_{\mathcal{D}}(E, \bigoplus X_n) \to \mathrm{Hom}_{\mathcal{D}}(E, \bigoplus X_n) \to \mathrm{Hom}_{\mathcal{D}}(E, X) \to 0$$

for all $E \in \mathcal{E}$.

Let $E \in \mathcal{E}$ and let $f : E \to X$ be a morphism. By the previous paragraph, we may choose $\alpha : E \to \bigoplus X_n$ lifting f. Then by assumption (2) we obtain a factorization $\alpha = (\bigoplus \beta_n) \circ \gamma$. For each n there is a morphism $\delta_n : E_n \to X_1$ such that δ_n and β_n map to the same element of $H(E_n)$. Then the compositions

$$E_n \to X_n \to X_{n+1} \quad \text{and} \quad E_n \to X_1 \to X_{n+1}$$

are equal by construction of $X_n \to X_{n+1}$. It follows that

$$\bigoplus E_n \to \bigoplus X_n \to X \quad \text{and} \quad \bigoplus E_n \to \bigoplus X_1 \to X$$

are the same too. Observing that $\bigoplus X_1 \to X$ factors as $\bigoplus X_1 \to X_1 \to X$, we conclude that

$$\mathrm{Hom}_{\mathcal{D}}(E, X_1) \to \mathrm{Hom}_{\mathcal{D}}(E, X)$$

is surjective. Since by construction the map $\mathrm{Hom}_{\mathcal{D}}(E, X_1) \to H(E)$ is surjective and by construction the kernel of this map is annihilated by $\mathrm{Hom}_{\mathcal{D}}(E, X_1) \to \mathrm{Hom}_{\mathcal{D}}(E, X)$ we conclude that $\mathrm{Hom}_{\mathcal{D}}(E, X) \to H(E)$ is a bijection for all $E \in \mathcal{E}$.

To finish the proof, consider the subcategory

$$\mathcal{D}' = \{Y \in \mathrm{Ob}(\mathcal{D}) \mid \mathrm{Hom}_{\mathcal{D}}(Y[n], X) \to H(Y[n]) \text{ is an isomorphism for all } n\}$$

As $\mathrm{Hom}_{\mathcal{D}}(-, X) \to H$ is a transformation between cohomological functors, the subcategory $\mathcal{D}'$ is a strictly full, saturated, triangulated subcategory of $\mathcal{D}$ (details omitted; see proof of Lemma 05RD). Moreover, as both H and $\mathrm{Hom}_{\mathcal{D}}(-, X)$ transform direct sums into products, we see that direct sums of objects of $\mathcal{D}'$ are in $\mathcal{D}'$. Thus derived colimits of objects of $\mathcal{D}'$ are in $\mathcal{D}'$. Since $\mathcal{E}$ is preserved by shifts, we conclude that $\mathcal{E} \subset \mathrm{Ob}(\mathcal{D}')$ by the result of the previous paragraph. To finish the proof we have to show that $\mathcal{D}' = \mathcal{D}$.

Let Y be an object of $\mathcal{D}$ and set $H(-) = \mathrm{Hom}_{\mathcal{D}}(-, Y)$. Then H is a cohomological functor which transforms direct sums into products. By the construction in the first part of the proof we obtain a morphism $\mathrm{colim}\, X_n = X \to Y$ such that $\mathrm{Hom}_{\mathcal{D}}(E, X) \to \mathrm{Hom}_{\mathcal{D}}(E, Y)$ is bijective for all $E \in \mathcal{E}$. Then assumption (1) tells us that $X \to Y$ is an isomorphism! On the other hand, by construction $X_1, X_2, \ldots$ are in $\mathcal{D}'$ and so is X. Thus $Y \in \mathcal{D}'$ and the proof is complete. □